Vector and Parametric Equation of a Plane

Communication

2. a. State two direction vectors for the xz-coordinate plane.

b. What do all direction vectors for the xz-coordinate plane have in common?

Knowledge/ Understanding

3. State two direction vectors for each of the following planes.

a. $\vec{r} = (9, 5, 2) + s(-3, 5, 2) + t(-6, 1, 2)$

b. a plane parallel to the plane $x = 3 + 5s + t$

$y = -2 - 5s + 6t$

$z = -2 + 3s - 2t$

c. the plane containing the intersecting lines $\vec{r} = (6, 5, -2) + s(4, -2, 1)$ and $\vec{r} = (-10, -3, 1) + t(-1, 5, 2)$

Knowledge/ Understanding

4. State two points that lie on each of the following planes.

a. $\vec{r} = (9, 4, -3) + t(-2, 2, 1) + p(0, -2, 6)$

b. $\vec{r} = (0, 1, 0) + t(1, 0, -2) + p(0, 0, 4)$

c. $x = 3 + 5s + t$

$y = -2 - 5s + 6t$

$z = -2 + 3s - 2t$

d. the xz-plane

Knowledge/ Understanding

5. Write parametric equations for each of these planes.

a. $\vec{r} = (-4, -6, 3) + s(5, 2, 3) + t(-4, -6, 3)$

b. $\vec{r} = (0, 0, 1) + s(0, 2, 0) + t(3, 0, 0)$

c. the xz-plane

Knowledge/ Understanding

6. Write a vector equation for each of these planes.

a. $x = -4 + s + 3t$
$y = -1 + 3s - 4t$
$z = 3 + 4s - t$

b. $x = 7s$
$y = 4$
$z = -2t$

c. the xz-plane

7. Determine a vector equation of each of the following planes.

a. the plane through the point $(-4, 5, 1)$ parallel to the vectors $(-3, -5, 3)$ and $(2, -1, -5)$

b. the plane containing the two intersecting lines $\vec{r} = (4, 7, 3) + t(1, 4, 3)$ and $\vec{r} = (-1, -4, 6) + s(-1, -1, 3)$

c. the plane containing the line $\vec{r} = (-3, 4, 6) + t(-5, -2, 3)$ and the point $(8, 3, 5)$

d. the plane containing the two parallel lines $\vec{r} = (0, 1, 3) + t(-6, -3, 6)$ and $\vec{r} = (-4, 5, -4) + s(4, 2, -4)$

e. the plane containing the three points $(2, 6, -5)$, $(-3, 1, -4)$, and $(6, -2, 2)$.

8. Determine parametric equations of each of the following planes.
   a. the plane through the point $(7, -5, 2)$ parallel to the vectors $(4, -1, 1)$ and $(-3, 4, 4)$
   b. the plane containing the two intersecting lines $\vec{r} = (5, 4, 2) + t(4, -2, 1)$ and $\vec{r} = (7, 4, -7) + s(-3, 1, 4)$
   c. the plane containing the line $\vec{r} = (1, 3, -1) + t(2, 2, -5)$ and the point $(8, 3, 5)$
   d. the plane containing the two parallel lines $\vec{r} = (3, 2, 2) + t(-9, 6, -6)$ and $\vec{r} = (1, 6, -6) + s(6, -4, 4)$
   e. the plane containing the three points $(2, 6, -5)$, $(-3, 1, -4)$, and $(6, -2, 2)$

9. Determine the vector equation of each of the following planes.
   a. the plane parallel to the yz-plane containing the point $(6, 4, 2)$
   b. the plane containing the origin and the points $(3, 3, 3)$ and $(8, -1, -1)$
   c. the plane containing the x-axis and the point $(-1, -4, -7)$

10. a. Explain why the three points $(2, 3, -1)$, $(8, 5, -5)$, and $(-1, 2, 1)$ *do not* determine a plane.
    b. Explain why the line $\vec{r} = (4, 9, -3) + t(1, -4, 2)$ and the point $(8, -7, 5)$ *do not* determine a plane.

11. Find vector and parametric equations of the plane that contains the line $x = 7 - t$, $y = -2t$, $z = -7 + t$ and that does not intersect the z-axis.

12. Demonstrate that a plane with a vector equation of the form $\vec{r} = (a, b, c) + s(d, e, f) + t(a, b, c)$ passes through the origin.

Answers

2. a. $(1, 0, 0)$, $(4, 0, -3)$ **b.** y component is 0
3. a. $(-3, 5, 2)$, $(-6, 1, 2)$ **b.** $(5, -5, 3)$, $(1, 6, -2)$
c. $(4, -2, 1)$, $(-1, 5, 2)$ **4. a.** $(9, 4, -3)$, $(7, 4, 4)$
b. $(1, 1, 2)$, $(1, 1, -2)$ **c.** $(3, -2, -2)$, $(9, -1, -1)$
d. $(5, 0, 1)$, $(-3, 0, 2)$ **5. a.** $x = -4 + 5s - 4t$
$y = -6 + 2s - 6t$, $z = 3 + 3s + 3t$
b. $x = 3t$, $y = 2s$, $z = 1$ **c.** $x = s$, $y = 0$, $z = t$
6. a. $\vec{r} = (-4, -1, 3) + s(1, 3, 4) + t(3, -4, -1)$
b. $\vec{r} = (0, 4, 0) + s(7, 0, 0) + t(0, 0, -2)$
c. $\vec{r} = s(1, 0, 0) + t(0, 0, 1)$
7. a. $\vec{r} = (-4, 5, 1) + s(-3, -5, 3) + t(2, -1, -5)$
b. $\vec{r} = (4, 7, 3) + s(1, 4, 3) + t(-1, -1, 3)$
c. $\vec{r} = (8, 3, 5) + s(5, 2, -3) + t(11, -1, -1)$
d. $\vec{r} = (0, 1, 3) + s(2, 1, -2) + t(4, -4, 7)$
e. $\vec{r} = (2, 6, -5) + s(5, 5, -1) + t(4, -8, 7)$

8. a. $x = 7 + 4s - 3t$, $y = -5 - s + 4t$, $z = 2 + s + 4t$
b. $x = 5 + 2s + 4t$, $y = 4 - 2t$, $z = 2 - 9s + t$
c. $x = 8 + 5s + 2t$, $y = 3 - 2s + 2t$, $z = 5 + 11s - 5t$
d. $x = 3 + s + 3t$, $y = 2 - 2s - 2t$, $z = 2 + 4s + 2t$
e. $x = 2 + 5s + 4t$, $y = 6 + 5s - 8t$, $z = -5 - s + 7t$
9. a. $\vec{r} = (6, 4, 2) + s(0, 1, 0) + t(0, 0, 1)$
b. $\vec{r} = s(1, 1, 1) + t(8, -1, -1)$ **c.** $\vec{r} = s(1, 0, 0) + t(1, 4, 7)$
10. a. the three points are collinear **b.** the point is on the line
11. $\vec{r} = (7, 0, -7) + s(0, 0, 1) + t(1, 2, -1)$; $x = 7 + t$, $y = 2t$, $z = -7 + s - t$ **13. b.** All points in and on the parallelogram